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Direct and configurational paths of knowledge-oriented leadership, entrepreneurial orientation, and knowledge management processes to project success

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Abstract

Purpose – Drawing on the knowledge-based view, the study aims to investigate the impact of knowledge management enablers (entrepreneurial orientation and knowledge-oriented leadership) on knowledge management processes and project success. The study further ascertains the specific combinations of knowledge management enablers and knowledge management processes that can lead to project success.

Design/methodology/approach – Survey data were collected from 222 project workers in information technology projects, and the proposed relationships were assessed through partial least squares structural equation modeling while configuration paths were assessed using fuzzy-set qualitative comparative analysis.

Findings – The study found a significant impact of entrepreneurial orientation and knowledge-oriented leadership on knowledge management processes and project success. The analysis also revealed that knowledge management processes did not significantly impact project success. Moreover, the insights from fuzzy-set qualitative comparative analysis show a clear pattern of equifinality, in that there are multiple combinations of knowledge management enablers and knowledge management processes that can lead to a successful project.

Originality/value – The current study is one of the earlier studies to provide insights to knowledge-based view by demonstrating the inter-relationship of entrepreneurial orientation and knowledge-oriented leadership with knowledge management processes and project success. To the best of authors' knowledge, this is the first study to assess the impact of knowledge-oriented leadership on project success. With limited studies on impact of entrepreneurial orientation and knowledge-oriented leadership on knowledge management processes, the study enriches the literature on linkage of entrepreneurial orientation and knowledge-oriented leadership with knowledge management processes. Methodological contributions include use of fuzzy-set qualitative comparative analysis to reveal multiple pathways to project success.

Keywords Entrepreneurial orientation, Knowledge-oriented leadership, Knowledge management processes, Project success, fuzzy-set qualitative comparative analysis

Paper type Research paper

Introduction

To achieve competitive advantage, firms need to combine and deploy several firm-level resources and capabilities (Ferraris *et al.*, 2019). Knowledge has been identified as one of the key resources that can help improve the human capital for different types of organizations (Bontis and Nikitopoulos, 2001). Knowledge management (KM) is seen as the management processes and activities that an organization practices to improve the effectiveness of generating, creating and sustaining organizational intellectual assets (Ramadan *et al.*, 2017).

KM enablers and KM processes are described as KM capability (Iqbal *et al.*, 2019). KM enablers include factors like culture, leadership and intellectual capital that enable KM

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processes (Iqbal *et al.*, 2019; Obeidat *et al.*, 2017). On the other hand, KM processes include practices pertinent to knowledge acquisition, creation, sharing and utilization that foster sustainable competitive advantage (Torres *et al.*, 2018). KM is inevitable for project-based organizations. Handzic *et al.* (2016) have noted that project-based organizations are facing the challenge of identifying the project-related knowledge assets that what they possess. They also confront challenges to identify how they can improve the Project Success (PS). With much of the existing literature focused on the role of KM in improving organization performance, there is little focus on the role of KM in PS (Gasik, 2015). An important theme in the project management research literature are success factors that impact PS (Jitpaiboon *et al.*, 2019). Research has asked for identification of different factors that can be critical in success of a project (Aga *et al.*, 2016). Existing literature has identified several gaps pertinent to the role of KM enablers (entrepreneurial orientation (EO) and knowledge-oriented leadership (KoL)), KM processes and PS that shall be addressed.

First, much attention in the entrepreneurship literature has been on organizational performance (Engelen *et al.*, 2015; Al-Dhaafri *et al.*, 2016). However, very few studies assess the relationship between entrepreneurship and PS (Kuura *et al.*, 2014). Martes *et al.* (2015) assessed the impact of EO on project management maturity and found a positive relationship. More recently, Martens *et al.* (2018) in the first attempt to ascertain the direct impact of EO on PS found a significant relationship.

Second, to the best of authors' knowledge and with search on the Web of Science, there is only one research that included the variables knowledge leadership and PS in a single study (see Yang *et al.*, 2014). The study by Yang *et al.* (2014) did not assess the impact of KoL on project performance, rather assessed the impact of KoL on customer KM. Hence, our study is among the first to consider knowledge leadership as an important antecedent of PS.

Third, KM processes, in particular, knowledge sharing has captured a greater part of scholarly attention (Masa'deh *et al.*, 2017). The role EO has been assessed as a predictor of a limited number of KM processes such as knowledge utilization (Wach *et al.*, 2018), knowledge sharing (Hormiga *et al.*, 2017) and knowledge creation (Jiang *et al.*, 2019). Additionally, EO literature linkage to knowledge-related activities in the organization has attracted much less scholarship (Jiang *et al.*, 2019). Hence, the present study has taken into consideration a total of seven KM processes that include knowledge identification (KI), knowledge creation (KC), knowledge collection (KL), knowledge organizing (KO), knowledge storage (KS), knowledge sharing (KD) and knowledge application (KA).

Fourth, the research on KoL is still in its infancy (Shamim *et al.*, 2019). Donate and de Pablo (2015) assessed the impact of KoL on four KM processes and have stressed the necessity to carry out additional research on the role of KoL in improving KM processes. Additionally, there is significantly limited research on the impact of leadership on knowledge-related practices (Bavik *et al.*, 2018). Furthermore, existing research has evaluated the impact of limited number of KM processes on PS (see Yang *et al.*, 2012; Fedor *et al.*, 2003).

PS can be affected by a number of different factors (Aga *et al.*, 2016), and a combination of different factors could enhance the chances of PS. However, prior research has relied primarily on symmetrical correlational methods to provide general information about the factors that affect PS, producing a single dominant "net effects" model. Little research has focused on the combinations of causally relevant factors (Li, 2019). Fuzzy-set qualitative comparative analysis (fsQCA) is an inductive, iterative method that reveals patterns in the data that tend to be obscured by symmetric methods (Ragin and Fiss, 2008). While fsQCA is becoming increasingly common in organization and management science for analyzing an intermediate number of cases from a set-theoretic perspective, fsQCA is quite novel to the research on project management (Verweij, 2015). This method allows examining configurations of conditions in relation to a particular outcome, in a way that is not possible by means of a linear additive approach. In the instances where the interplay between

conditions (i.e. the configuration) is believed to be of central importance, fsQCA offers more accurate predictions of the outcome relative to the linear additive approach (de Rooij *et al.*, 2019). The set-theoretic approaches allow studying how factors combine into configurations of necessary and sufficient conditions underlying outcomes (Ragin and Fiss, 2008). Because projects consist of many interrelated parts, a type of analysis that is sensitive to the potential complexity hereof (like fsQCA) seems particularly promising to study KM in project management literature.

The theoretical lens for the study is knowledge-based view (KBV) (Grant, 1996), which argues that organizations are social entities that use and store internal knowledge, competencies and capabilities that are vital for the firm's survival, growth and success. KBV has been used extensively in the KM literature (e.g. Iqbal *et al.*, 2019; Shahzad *et al.*, 2016; Naqshbandi and Jasimuddin, 2018).

Based on the aforementioned literature gaps, the research aims to ascertain the inter-relationship between KM enablers (KoL and EO), KM processes and PS. Furthermore, the study aims to identify the configuration/causal recipes that could lead to successful projects through combination of KM and KM processes. The research makes several significant contributions. Through the integration of KBV, we identify how KM enablers and different KM processes are associated with PS, extending the limited research on the linkage between EO, KoL and PS. The research would enrich the KM literature and help understand the role of KM in fostering PS. With limited studies on the impact of EO on limited number of KM processes, the study would help enrich the literature on the linkage between EO and KM processes. Similarly, the impact of knowledge leadership has been studied on four KM processes (see Shamim *et al.*, 2019); the present study focuses on seven KM processes. With knowledge identified as critical to managing intellectual capital, intellectual capital is known as an intangible asset that corresponds to knowledge that creates value for a firm (Shahzad *et al.*, 2019); the study would help identify the factors that can help foster KM processes that can be critical improving the intellectual capital within an organization. Methodological contributions include the use of fsQCA to reveal multiple pathways to PS and expose asymmetric relationships between PS and some of its antecedents. The study contributes toward an existing trend observed in other fields, such as marketing or management (Fiss, 2011), to combine symmetric and asymmetric statistical tools to further develop our understanding of organizational issues. For instance, we add to existing organizational studies in providing with fsQCA the appropriate analytical tool to better understand contingencies and interactions that might not be immediately obvious from the use of traditional symmetric methods alone (Cabrilo and Dahms, 2018). Finally, we empirically test our model in the Pakistani business context because little work has so far has been done in developing economies.

Literature review and hypotheses development

Knowledge management (KM) enablers

KBV contends that knowledge is the key strategic asset and resource for the organization that enables them to create value (Grant, 1996). KM enablers are factors or institutional mechanisms that stimulate KC, facilitate KD and promote knowledge transfer (Ramjeawon and Rowley, 2017). Recently, there has been an increasing focus on ascertaining the relationship of KM enablers with KM (Santoro *et al.*, 2018; Ramachandran, 2018; Payal *et al.*, 2019), and there have been increasing call in existing research on assessing the role of KM enablers in fostering KM processes (Santoro *et al.*, 2018; Iqbal *et al.*, 2019). The present study considers two enablers, namely, EO and KoL.

EO is a strategic intent of an organization and relates to basic policies and practices for the development of entrepreneurial actions that help attain competitive advantage (Martens *et al.*, 2018). Conceptually, EO is framed as strategic position of firm toward entrepreneurship

development that will help to find new opportunities and allow to take optimal level of benefits from it (Anderson *et al.*, 2015). EO has usually been viewed as the singular quality that risk-taking, innovative and proactive behaviors have in common (Jiang *et al.*, 2019).

Leadership has been identified as one of the enablers of KM (Koohang *et al.*, 2017; Iqbal *et al.*, 2019). In the context of KM, leadership is regarded as the ability of an organization to configure KM behaviors with organizational strategy, identify knowledge opportunities, encourage KM values and promote learning in the organization (Koohang *et al.*, 2017). KoL can exploit knowledge among the employees (Shariq *et al.*, 2019). The concept of knowledge leadership has been recently developed (Shamim *et al.*, 2019; Donate and de Pablo, 2015), and conceptualization of knowledge leadership and KM in the project management literature is still rudimentary (Yang *et al.*, 2014). KoL can be defined as joint or individual attitude or action, observed or charged that stimulates some latest and the most important knowledge to be shared, created and used in such a way that brings change in the thinking and collective outcome (Mabey *et al.*, 2012). Knowledge leadership is imperative for organizations to enhance their performance through the effective development and implementation of KM initiatives (Donate and de Pablo, 2015). A knowledge-oriented leader shall act as a role model, promote learning by motivating and empowering the followers intellectually, institutionalize learning through the provision of incentives and training, develop a learning culture that endures mistakes, boost discipline and develop KS, knowledge transfer and KA mechanisms (Naqshbandi and Jasimuddin, 2018).

Knowledge management processes

The goal of the KM processes is to make an organization aware of its knowledge at the individual and collective level and utilize that knowledge to shape itself and make its business process efficient and effective (Azan *et al.*, 2017). In this research study, seven KM processes are discussed (Masa'deh *et al.*, 2017) i.e. identification, creation, collection, organizing, storage, dissemination and application.

KI encompasses an organization's steps to identify the relevant and needed knowledge that exists within their boundaries (Masa'deh *et al.*, 2017). *KC* is the ability of an organization to create new knowledge within the context of the organization and throughout all organizational levels and to incorporate its outcomes in the organization (Yang *et al.*, 2010). *KL* is a process by which needed knowledge is gained by consulting and by buying from other sources (Gold *et al.*, 2001). *KO* refers to describing, representing, filing and organizing (bringing together) knowledge (Hjørland, 2008). *KS* can be defined as the ability of an organization to store the new knowledge in organizational memory so that others can access and utilize it (Allameh *et al.*, 2011). *Knowledge dissemination* can be defined as the process by which knowledge is shared among people. It involves the exchange skills, knowledge and experiences with the organization (Lin, 2007). *KA* is the process of utilizing available knowledge for achieving organizational goals (Asoh *et al.*, 2007).

Project success

Project is a set of inter-related activities with team structure often consisting of different group of people with different level of skillsets and specialties, working together for gaining the organization's goal and plan to complete their task in specific time and cost (Sydow *et al.*, 2004). Success is defined as completing the particular project or specific activity with the help of individuals, teams or groups with optimal use of resource (Basten *et al.*, 2011). PS covers the broad objectives, the value that project promises to deliver with all its benefits, (DeWitt, 1998). PS has been a central topic in project management literature (De Witt, 1998). To better understand PS, Turner (2009) suggested that project needs to be defined in terms of PS factors. PS factors are elements of a project which, when influenced, increase the probability of success (Jugdev and Muller, 2005).

KM enablers and KM processes

EO and KM processes. Entrepreneurship is about taking risk and is linked to fundamental policies and practices for the development of entrepreneurial actions leading to competitive advantage (Martens *et al.*, 2018). From a KM perspective, an organization's system that fosters experimentation and risk-taking aids learning, KC and dissemination (Stuetzer *et al.*, 2018). Conceptually, EO is utilized by decision makers to provide their organization with a purpose, support vision and create competitive advantages (de Guimarães *et al.*, 2018). These rules and directives (procedures, plans, policies and practices) have been described as a key mechanism for integrating knowledge held by individuals (Grant, 1996), which, according to KBV, are used by the organizations for its survival, growth and success (Hakanson, 2010). The study argues that organizations that are focused on taking risks and on policies for development will have focus on nurturing KM processes. This holds true in light of KBV because the issues with which the KBV concerns itself include the organizational structure, the role of management, the allocation of decision-making rights and innovation (Grant, 1996). EO has been identified to have a positive effect on KC (Jiang *et al.*, 2019) and knowledge acquisition (Gupta and Moesel, 2007). Hence, based on the aforementioned literature, it is proposed that:

H1. EO has a significant impact on KM processes.

Knowledge-oriented leadership and KM processes. Without the commitment of leadership to KM activities, it is difficult to achieve the success of KM (Civi, 2000). Knowledge-oriented leaders create the thirst for knowledge through provision of a clear and inspiring vision and providing direction for future endeavors (Shamim *et al.*, 2019). In general, knowledge leadership behavior should propel KM initiatives of creation, storage, transfer and application (Donate and de Pablo, 2015). According to KBV, knowledge is created, stored and utilized by individuals and not by organizations as a whole, coordinating and integrating the knowledge held by individuals is a difficult task (Miles, 2012). This can be made possible when an organization has knowledge experts in managerial positions who are competent and know how to acquire, store and transfer knowledge. Knowledge-oriented leaders are the essential unit of overall knowledge process of firms who act as role models, facilitators, motivators and advisors in promoting successful use of knowledge (Naqshbandi and Jasimuddin, 2018). According to the absorptive capacity theory, knowledgeable people can adapt and modify the new knowledge for the organization (Sjödin *et al.*, 2019). The present study argues that knowledge leaders would be more absorbent and would cater for initiatives that would facilitate the KM processes. They can provide an environment that empowers employees to utilize their KM skills, gain knowledge from organizational resources and contribute to organizational knowledge (Abualoush *et al.*, 2018). Hence, based on the aforementioned literature, it is proposed that:

H2. Knowledge leadership has a significant impact on KM processes.

KM enablers and project success

Entrepreneurial orientation and project success. Entrepreneurship is a well-known path to the achievement of growth and success (Del Giudice *et al.*, 2019). Traditionally, the literature on entrepreneurship has focused on the relationship between EO and organizational performance (Engelen *et al.*, 2015), while for about 30 years, organizational performance found to be one of the most investigated/studied areas in EO (Martens *et al.*, 2018). The fundamental argument is that "EO may be viewed as the entrepreneurial strategy-making processes that key decision makers use to enact their firm's organizational purpose, sustain its vision, and create competitive advantage(s)" (Rauch *et al.*, 2009, p. 762). The evidence from the existing research that EO leads to a superior organizational performance refers to the possibility that EO can contribute to a

successful project. In an attempt to understand factors affecting PS, Vezzoni *et al.* (2013) noted that the two most significant PS factors were preparing to face risks and empowerment, which can be related to the EO characteristics of risk-taking and autonomy (Martens *et al.*, 2018). This shows that with higher EO, a business can undertake more innovative, risky and uncertain projects (Wales, 2016). Raziq *et al.* (2018) noted that proactivity and autonomy of project managers are important factors that may affect PS. Additionally, there has only been a single study on the linkage between EO and PS, Martens *et al.* (2018) found that organizations that has EO have an increased likelihood of a of a successful project. Hence, based on the aforementioned literature, it is proposed that:

H3. There is a significant impact of EO on PS.

Knowledge leadership and project success. Leadership acts as the foundation of the elevated trust within the organization, the effective KM processes and the superior organizational performance (Koohang *et al.*, 2017). Shahzad *et al.* (2016) argued the importance of an effectively managed system for KM and stressed that only the presence of knowledge cannot guarantee sustainable competitive advantage and organizational success. KBV identifies knowledge as a key resource and a strategic asset that empowers the business to create value and help members share information and turn this sharing into actions that improves organizational effectiveness (Grant, 1996; Ferraris *et al.*, 2017). In project-based organizations, effective leadership improves the commitment by instilling a positive attitude and culture that aids in achievement of PS (Aga *et al.*, 2016). Project managers' competencies are vital in improving project performance (Maqbool *et al.*, 2017). Some knowledge, skills and abilities have emerged as relevant to PS; these include critical thinking, flexibility, participation, development, documentation, implementation, quality assurance, communication, project reviews and leadership (Gallagher *et al.*, 2015). Leadership that inspires, enhances employees' knowledge and learning to encourage them to be innovative with problem-solving tools provide intellectual stimulation and inspire the employees, enhance communication, trust and the sharing of knowledge is expected to promote higher performance (Para-González *et al.*, 2018). Hence, based on the aforementioned literature, it is proposed that:

H4. There is a significant impact of knowledge leadership on PS.

Knowledge management processes and project success

Knowledge as a vital strategic resource for modern business is widely recognized (Santoro *et al.*, 2019), as it plays a critical role in generating competitive advantage (Ferraris *et al.*, 2018). According to KBV, the processes that organizations utilize to access and utilize their knowledge for value extraction and creation become a lynchpin for performance improvement (Hsu and Sabherwal, 2011). KBV supports the idea that when knowledge is effectively managed, it creates distinguishing capabilities which contribute to improved performance (Leal-Rodríguez *et al.*, 2013). There is an increasing need for the use of KM as an enabler to improve organization outcomes (Choi *et al.*, 2019). With organization facing rapid changes both within and outside the organization, knowledge-related capabilities can provide significant support in sustaining competitive advantage and be successful (Torres *et al.*, 2018; Shahzad *et al.*, 2016). For example, KM processes like acquisition, sharing and utilization not only support learning and decision-making, but also drives productivity and profitability through innovative ideas and novelty (Chiu and Chen, 2016). In light of KBV, successful utilization of knowledge-based resources and effective implementation of KM processes is imperative to achieve better performance (Mazdeh and Hesamamiri, 2014). Several studies have identified KM as a mean to improve organizational performance (Iqbal *et al.*, 2019; Shahzad *et al.*, 2016); very few published empirical studies, only three found in the Web of Science (see Fedor *et al.*, 2003; Yang *et al.*, 2012; Yang *et al.*, 2017) have focused on the direct

effects of KM practice on the performance of projects. While Yang *et al.* (2012) and Yang *et al.* (2017) found a significant impact of KM on PS, the relationship was moderately significant in the study of Fedor *et al.* (2003). Hence, there is lack of conclusive evidence on the linkage between the two concepts. Hence, based on the aforementioned literature, it is proposed that:

H5. KM processes have a significant impact on PS.

Equifinality

The discussion so far showed that there is a strong interrelationship between the concepts covered in this study. This indicates that several ways to achieve PS in the context of KM enablers and KM processes may exist. Research studies also show that the relationship between leadership and KM in project management is limited and requires further attention (Todorović *et al.*, 2015). The research on PS has noted that success is a complex concept (Besner and Hobbs, 2006) and what leads to PS is not straightforward. A limitation of existing project management research on PS has been lack of investigation into the inter-relationship between factors that could affect PS (Ika *et al.*, 2012). This complex interrelationship hints toward equifinality (Fiss, 2011). That means there are several ways to achieve PS. This suggests the existence of complex configurations of KM enablers and KM processes with PS. In line with this perspective, this study posits that:

H6. Varied combinations of KM enablers (EO and KoL) and KM processes (KI, KC, KL, KO, KS, KD and KA) are associated with PS.

Research framework

The research framework for the study is represented in Figure 1.

Research methodology

Data and procedure

To ascertain the direct and configurational paths from KM enablers and KM processes to PS, the research setting was IT sector. As IT companies probably have a broader spectrum of KM strategies, they place a greater emphasis on the management of knowledge (Donate and de Pablo, 2015). In addition, knowledge organizations from technology-intensive industries require a different management approach from non-knowledge organizations, and the role of leadership and HR management should be distinctive and supportive of KM processes (Yahya and Goh, 2002). Data were collected from project workers through personal visits using convenience sampling. A total of 350 questionnaires were distributed; 251 questionnaires were received, reflecting a response rate of 71.71 percent. A total of 29 questionnaire were incomplete or invalid and were discarded; 222 questionnaires were found usable for statistical analysis. The questionnaire used a seven-point Likert scale from "1" referring to "strongly disagree" to "7" referring to "strongly agree".

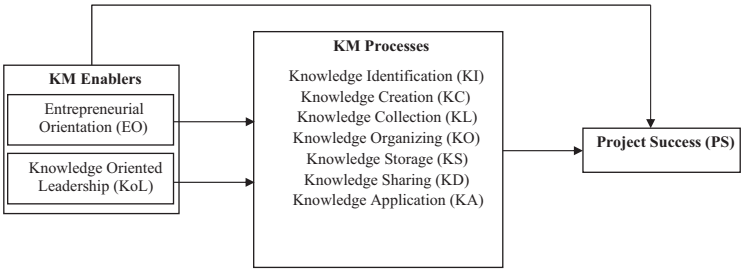


Figure 1.
Conceptual framework

Measures

The seven-item scale for EO is based on Tuan (2017). The six-item scale on KoL was adapted from Donate and de Pablo (2015). The KM processes scale was adapted from Masa'deh *et al.*, (2017) with 28 items. The six-item scale measuring PS was adapted from Engelbrecht *et al.* (2017). The final questionnaire is presented in Table A1.

Data analysis and results

To substantiate the direct relationships (*H1–H5*), the study used the partial least squares structural equation modeling (PLS–SEM) path model. We first tested the measurement model to explore the reliability and validity. Subsequently, the structural model was tested to examine the hypothesized relationships among the constructs. This study used fsQCA to address H6 by identifying configurations of KM enablers and KM processes for achieving PS.

Measurement model assessment

As part of the measurement model evaluation, none of the items was removed from the analysis because of low factor loadings (<0.600) (Gefen and Straub, 2005). To test the reliability of the constructs, the study used Cronbach's alpha and composite reliability (CR). All the CRs were higher than the recommended value of 0.700 (Wasko and Faraj, 2005). Cronbach's alpha of each construct exceeded the 0.700 threshold. Convergent validity was acceptable because the average variance extracted (AVE) was over 0.500. The results for reliability and validity along with the factor loadings for the items are presented in Table I. Discriminant validity was assessed by comparing the correlations among the latent variables with the square root of AVE (Fornell and Larcker, 1981); this is shown in Table II.

Structural equation model

The R^2 values of KI (0.238), KC (0.349), KL (0.328), KO (0.312), KS (0.344), KD (0.339), KA (0.334) and PS (0.504) support the model's in-sample predictive power (Sarstedt *et al.*, 2014). Similarly, the results from blindfolding with an omission distance of seven yield Q^2 values well above zero (Table III), thus supporting the model's predictive relevance in terms of out-of-sample prediction (Hair *et al.*, 2014). Further analysis of the composite-based standardized root mean square residual (SRMR) yields a value of 0.06, which confirms the overall fit of the PLS–SEM (Hair *et al.*, 2014).

Applying the bootstrapping procedure (5,000 bootstrap samples) provides the *p*-values (Table III). The empirical results support the majority of hypothesized path model relationships among the constructs. Most notably, EO and KoL had a significant impact on the KM processes and PS. However, contrary to the hypotheses, KM processes play no role in improving PS.

fsQCA approach

The study uses fsQCA to analyze the aforementioned antecedent conditions (EO, KoL, and seven KM Processes) explaining PS.

Calibration

The first step of fsQCA is to calibrate the data, which we did in accordance with the procedure described by Ragin (2008). The fuzzy-set scores range from 0.00 to 1.00, with 0.00 for full non-membership and 1 (in most cases, 0.95) for full membership. The mid-point 0.50 is treated as the crossover point. Table IV shows the data statistics and calibration cut-off points chosen.

	Λ	α	CR	AVE
EO		0.821	0.867	0.482
EO1	0.696			
EO2	0.664			
EO3	0.661			
EO4	0.731			
EO5	0.742			
EO6	0.684			
EO7	0.676			
KoL		0.849	0.888	0.571
KOL1	0.741			
KOL2	0.768			
KOL3	0.813			
KOL4	0.773			
KOL5	0.755			
KOL6	0.677			
KM processes				
KI		0.82	0.881	0.651
KI1	0.766			
KI2	0.879			
KI3	0.828			
KI4	0.747			
KC		0.809	0.875	0.636
KC1	0.731			
KC2	0.808			
KC3	0.834			
KC4	0.814			
KL		0.799	0.868	0.622
KL1	0.811			
KL2	0.805			
KL3	0.75			
KL4	0.788			
KO		0.779	0.857	0.6
KO1	0.73			
KO2	0.723			
KO3	0.8			
KO4	0.839			
KS		0.722	0.828	0.547
KS1	0.686			
KS2	0.776			
KS3	0.79			
KS4	0.701			
KD		0.787	0.861	0.609
KD1	0.757			
KD2	0.812			
KD3	0.748			
KD4	0.802			
KA		0.794	0.866	0.617
KA1	0.788			
KA2	0.804			

(*continued*)

Table I.
Items loadings, reliability and convergent validity

	Λ	α	CR	AVE
KA3	0.779			
KA4	0.772			
PS		0.855	0.892	0.58
PS1	0.785			
PS2	0.771			
PS3	0.736			
PS4	0.774			
PS5	0.711			
PS6	0.79			

Table I.

	EO	KoL	KI	KC	KL	KO	KS	KD	KA	PS
EO	*0.694*									
KoL	0.586	*0.756*								
KI	0.468	0.388	*0.807*							
KC	0.520	0.532	0.598	*0.798*						
KL	0.529	0.489	0.592	0.711	*0.789*					
KO	0.488	0.506	0.616	0.604	0.625	*0.774*				
KS	0.515	0.530	0.466	0.496	0.533	0.614	*0.740*			
KD	0.551	0.477	0.524	0.600	0.631	0.564	0.563	*0.780*		
KA	0.550	0.467	0.473	0.510	0.497	0.551	0.450	0.665	*0.786*	
PS	0.591	0.628	0.381	0.488	0.501	0.482	0.521	0.481	0.454	*0.762*

Table II.
Discriminant validity

Note(s): Statistics along the diagonal in italics represent the square root of AVE

Necessity conditions

The next step is the identification of the necessary conditions for the outcome. Necessary condition analysis assesses if a single condition is always present, or absent, when the outcome is present (or absent). Ragin (2008) proposes 0.900 consistency for causal factors that by itself can explain the outcome (PS). The consistency scores ranged from 0.75 to 0.82 (Table V). The superscript indicates those conditions that meet, or exceed, the 0.65 consistency benchmark. The results (Table V) show that consistency values are greater than 0.65 for all causal conditions. Hence, all causal conditions are usually necessary conditions, but none of our conditions is able to explain PS by itself. Conversely, consistency values for negated conditions (~) are all less than or close to 0.500; this shows that they are more often than not a necessary condition.

Solution

Table VI presents fsQCA analysis results for PS using notations of Ragin and Fiss (2008). The overall solution as well as each part of the overall solution has a consistency greater than 0.80, and the overall solution consistency was 0.84. As these three solutions have a distinguishing combination of conditions, the existence of multiple solutions sufficient for the PS points to equifinality (Fiss, 2011), supporting *H6*. The fsQCA analysis reveals that for PS, KL and KS were present in two of the three configurations. The results confirm that fsQCA accepts variables to be causally related in one configuration, yet unrelated in others. Regarding the coverage values, an overall solution coverage score of 0.76 means that the three configurations of causal conditions "explain" 76 percent of PS falling within the suggested limit of 0.25–0.90 (Ragin, 2008).

	Path coefficient	Standard deviation	*t*-statistics	*p*-values	Result
EO → KA	0.421	0.086	4.889	0.000	H1 is supported
EO → KC	0.317	0.075	4.210	0.000	
EO → KD	0.414	0.078	5.290	0.000	
EO → KI	0.366	0.089	4.109	0.000	
EO → KL	0.369	0.085	4.321	0.000	
EO → KO	0.292	0.084	3.486	0.000	
EO → KS	0.310	0.077	4.006	0.000	
KoL → KA	0.219	0.075	2.921	0.004	H2 is supported
KoL → KC	0.346	0.080	4.335	0.000	
KoL → KD	0.234	0.078	3.000	0.003	
KoL → KI	0.173	0.083	2.074	0.038	
KoL → KL	0.273	0.083	3.281	0.001	
KoL → KO	0.335	0.082	4.082	0.000	
KoL → KS	0.348	0.072	4.818	0.000	
EO → PS	0.237	0.094	2.533	0.011	H3 is supported
KoL → PS	0.330	0.088	3.750	0.000	H4 is supported
KA → PS	0.036	0.095	0.382	0.702	H5 is not supported
KC → PS	0.029	0.097	0.295	0.768	
KD → PS	0.018	0.100	0.182	0.855	
KI → PS	−0.044	0.088	0.500	0.617	
KL → PS	0.098	0.093	1.057	0.291	
KO → PS	0.038	0.100	0.378	0.705	
KS → PS	0.129	0.087	1.477	0.140	
R^2 KI = 0.238	Q^2 KI = 0.142				
R^2 KC = 0.349	Q^2 KC = 0.206				
R^2 KL = 0.328	Q^2 KL = 0.187				
R^2 KO = 0.312	Q^2 KO = 0.167				
R^2 KS = 0.344	Q^2 KS = 0.176				
R^2 KD = 0.339	Q^2 KD = 0.190				
R^2 KA = 0.334	Q^2 KA = 0.189				
R^2 PS = 0.504	Q^2 PS = 0.267				

Table III.
Direct relationship hypotheses results

Variables	Mean	SD	Min	Max	Calibration (0.95, 0.50, 0.05)
KoL	4.83	1.17	1.67	7	6.67, 5, 2.67
EO	4.60	1.10	1.86	7	6.14, 4.71, 2.71
KI	4.48	1.32	1	7	6.25, 4.75, 2
KC	4.47	1.33	1.25	7	6.75, 4.50, 2
KL	4.46	1.23	1	7	6.5, 4.75, 2.25
KO	4.48	1.20	1	6.5	6.25, 4.5, 2.25
KS	4.54	1.18	1.25	7	6.5, 4.75, 2.5
KD	4.48	1.29	1	7	6.5, 4.5, 2.25
KA	4.67	1.18	1	7	6.25, 4.75, 2.5
PS	4.86	1.12	1.83	7	6.67, 5, 2.83

Table IV.
Descriptive statistics and calibration cut-offs

Predictive validity

To perform predictive validity, the sample was split into two subsamples. fsQCA analysis was used to model Subsample 1. The results from Subsample 1 were compared to results from the overall sample. The results from the Subsample 1 showed a solution consistency (0.82). This is akin to the results from the overall sample. An example of the predictive validity test applied to indicate a high score in PS with a combination of the algorithm

Conditions	Outcome: PS	
	Consistency	Coverage
KIC	0.75083[a]	0.73147
~KIC	0.54141	0.54518
KCC	0.78006[a]	0.76043
~KCC	0.54528	0.54872
KLC	0.7559[a]	0.79015
~KLC	0.58318	0.54867
KOC	0.79685[a]	0.75022
~KOC	0.51485	0.53776
KSC	0.75968[a]	0.77982
~KSC	0.56769	0.54305
KDC	0.77462[a]	0.75309
~KDC	0.53403	0.5389
KAC	0.80238[a]	0.75125
~KAC	0.50249	0.52811
EOC	0.82571[a]	0.79868
~EOC	0.4894	0.49649
KoLC	0.82617[a]	0.81661
~KoLC	0.52315	0.51908

Table V.
Overview of the necessary conditions

Note(s): ~ indicates the absence of a condition.[a]Meets the 0.65 consistency benchmark for usually necessary conditions

Condition	Outcome: PS		
	1	2	3
KM enablers			
EO		•	
KoL	●		
KM processes			
KI		•	•
KC		•	•
KL		●	●
KO			•
KS	●		●
KD			•
KA			•
Raw coverage	0.69	0.59	0.52
Unique coverage	0.14	0.05	0.01
Consistency	0.87	0.90	0.93
Solution coverage 0.76			
Solution consistency 0.84			

Table VI.
Results of intermediate solution

Note(s): Large circles indicate core conditions, small circles indicate peripheral condition and blank space indicate do not care

(KI*KC*KL*KO*KS*KD*KA) results for Subsample 1 (consistency: 0.94 and coverage: 0.50) were found like the results from fsQCA in the whole sample (Table VI).

Then, as an example, a configural model (PS = KI*KC*KL*KO*KD*KA*EO) of Subsample 1 (consistency: 0.95 and coverage: 0.49) were tested using the holdout sample (Subsample 2). The results from Subsample 2 show almost similar asymmetric relationships (consistency: 0.92 and coverage: 0.57). This proves the ability of the model in predicting outcome conditions with different data sets. The XY plot (Figure 2) of the example model from

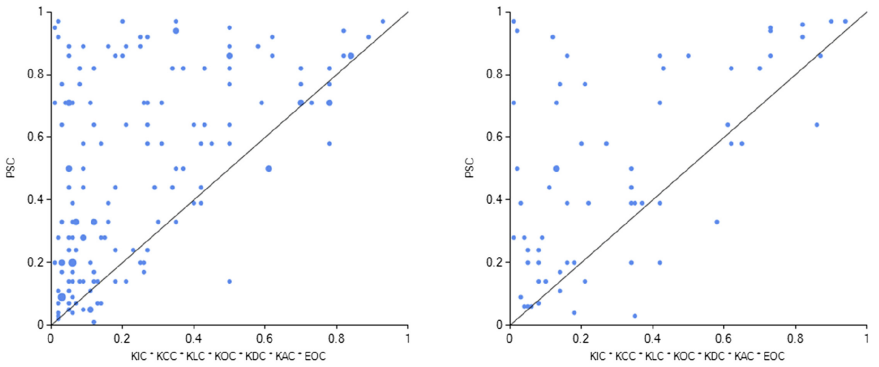


Figure 2.
XY plot for Subsamples 1 and 2

Subsamples 1 and 2 are provided. The models show that the subsample is causally relevant in predicting high levels of PS.

Discussion

The present study ascertained the inter-relationship between KM enablers, KM processes and PS. The study also assessed the configurational impact of KM enablers and KM processes on PS.

The study found a significant impact of EO on PS. The results of the study are similar to the findings of Martens *et al.* (2018). This shows that a project-based organization characterized by orientation toward research, innovativeness and risk will have a higher propensity for a successful project. The study also found a significant impact of KoL on PS. This shows that KoL behaves like a dynamic capability, with a focus on the continuous reconfiguration of a firm's knowledge position through initiatives to articulate, codify and utilize knowledge (Wang and Ahmed, 2007). A strong KoL position forces the firm to embark on substantial investment and development initiatives pertinent to KM (Donate and de Pablo, 2015).

Furthermore, the study also ascertained the significant impact of EO on KM processes. This is in line with the findings of Stuetzer *et al.* (2018) who found that management initiatives like experimentation and risk-taking, key ingredients of EO, affect how knowledge is created and shared. From a KM perspective, an organization's system that encourages risk-taking and experimentation would support learning as well as KC and dissemination (Mile, 2012). The study found a significant impact of KoL on KM processes. This is also true in light of existing research that found a significant impact of leadership on KM processes (Iqbal *et al.*, 2019). The role of KoL can be further emphasized in light of KBV, because coordinating and integrating knowledge held by diverse individuals is a difficult task for managers (Miles, 2012). A leader can help and improve this difficult situation by showing top management commitment and leadership in KM related activities (Naqshbandi and Jasimuddin, 2018; Civi, 2000).

Assessing the impact of KM processes on PS, the study found an insignificant impact of KM processes on PS. This is in contradiction of KBV that proposes that effective management of knowledge resources can facilitate organizations to achieve superior performance. According to KBV, the processes through which organizations access and utilize their knowledge for value extraction and creation become a lynchpin for performance improvement (Hsu and Sabherwal, 2011).

From the configurational analysis, we deduce that none of the conditions, i.e. neither KM enablers nor KM processes, is sufficient to explain PS on their own. Instead, our strongest solution for PS (Solution 1 in Table VI) indicates that for over half of the respondents in our sample, high KoL and KS lead to high PS. This showed that project workers deliver successful project through leadership that is focused on knowledge-related practices in combination with organizational ability to store knowledge. This finding expands previous studies (Donate and de Pablo, 2015), which found that improved performance outcomes are based on both KoL and KM processes. We not only show that KoL has a significant impact but also show that certain KM processes are more important than others. The presence of equifinality, which is slowly gaining momentum in the wider management literature (Fiss, 2011), can be found in our case, where combinations of KM enablers and KM processes lead to high level of PS.

Conclusion

This paper presents one of the earliest attempts to develop and test an integrated model that links KM enablers, namely, EO and KoL to KM processes and PS. This research adds to the KBV literature by exemplifying that EO and KoL can significantly enhance the KM processes, and that both EO and KoL have a significant role in improving the chances of a successful project. The data have been analyzed using symmetric PLS–SEM and asymmetric fsQCA technique. The SEM findings of this research render that EO and KoL are mandatory for successful implementation of KM processes and achieve PS. Moreover, the insights from our application of fsQCA technique show a clear pattern of equifinality. In particular, we show that there are multiple combinations of conditions that can lead to higher level of PS. Our study suggests that certain combinations of KM enablers and KM processes can be exploited to achieve high level of PS.

Implications

The research has significant implications. Theoretically, first, the study empirically validates the role of KoL and EO in enhancing KM processes. Second, it contributes to the stream of literature on leadership and entrepreneurship, highlighting the positive effect of developing EO and KoL on PS. Third, methodologically, the study contributed through the use of fsQCA to reveal multiple pathways to PS and expose asymmetric relationships between PS and some of its antecedents. Finally, a firm's ability to manage intellectual capital (IC) assets is inseparably linked to its KM capability. IC can be viewed as a knowledge stock that an organization possesses at a particular point in time, which is managed and accumulated through dynamic knowledge processes (Bontis, 2004). Hence, such knowledge-focused leadership and KM processes can be considered to be critical factors for facilitating the right conditions for a firm's IC to develop (Ramadan *et al.*, 2017). The current study focuses on factors that can foster human capital, which is the foremost component of IC that encapsulates knowledge-related practices and processes; the study helps ascertain the role of KoL and entrepreneurial leadership as critical factors that can foster human capital that through strengthening the KM practices. Focusing on these factors will help the organizations expand the knowledge base, which has the capacity to improve a firm's overall productivity. Focusing on the KM is increasingly important for IC as an enriched work context where workers are motivated and encouraged to transfer into the organization their unique individual knowledge is imperative for the sake of IC management. Organizations can only gain this individual human capital if the most talented employees will share their knowledge, skills and creativity with other workers. Nurturing IC in a knowledge-intensive context can also be significantly enhanced through looking at the firm as a cooperative

knowledge-sharing entity, i.e. investing in social capital. Social capital can be significantly enhanced through ensuring effective internal knowledge transfer and acquisition practices.

Furthermore, PS has been found to be a complex concept affected by different factors. The study further elucidates this notion and clarifies that combination of factors can help businesses improve their chance of PS. The findings guide managers to emphasize more on the KM enablers–KM processes combinations, to have an organization with KoL and EO that fosters KM processes working together to better understand synergistic effects that would ensure higher chances of PS. The study provides guidance for managers to capitalize on their leadership skills, improve EO and improve KM processes within the organization. Policymakers and managers should come up with a deliberate plan to foster EO and provide a supportive knowledge leadership to encourage, shape and sustain KM processes among their employees. Leadership or top management must promote entrepreneurial practices and must provide a clear vision about KM and share it with other members that KM processes are vital for success of the project.

Limitations and future research directions

This study is one of the few to combine symmetrical and non-symmetrical methods in the KM literature. Future studies could gain from also applying this approach. The study focused on information technology (IT) companies; future studies could apply the model in other industries. The data to test the model were collected from Pakistan. Future studies shall test the model in other developing and developed countries and may further compare the findings to assess the significance of the impact. The present study only assessed two KM enablers; future studies may asses the role of other KM enablers and shall also assess the impact of KM processes on IC.

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Appendix

EO
EO1: Our company strongly emphasizes R&D, technological leadership and innovations
EO2: Our company has introduced a lot of new lines of products or services in the past five years
EO3: The changes in product lines (types/number of products) for our company have usually been dramatic
EO4: Our company is typically the first to initiate actions to competitors, for which the competitors then respond
EO5: Our company has a strong propensity toward getting involved in high-risk projects (with a chance for high yield)
EO6: Owing to the nature of the environment, bold, wide-ranging acts are necessary to achieve the company's objectives
EO7: When confronted with decision-making situations involving uncertainty, our company typically adopts a cautious, "wait and see" posture to minimize the probability of making costly erroneous decisions (reverse-coded item)
KoL
KOL1: Leadership has been creating an environment for responsible employee behavior and teamwork
KOL2: Managers are used to assuming the role of knowledge leaders, which is mainly characterized by openness, tolerance of mistakes and mediation for the achievement of the firm's objectives
KOL3: Managers promote learning from experience, tolerating mistakes up to a certain point
KOL4: Managers behave as advisers, and controls are just an assessment of the accomplishment of objectives
KOL5: Managers promote the acquisition of external knowledge
KOL6: Managers reward employees who share and apply their knowledge
KM processes
KI
KI1: The company I work for has the available knowledge about the employees' skills
KI2: The company I work for tries to adjust the employees' knowledge and abilities with require knowledge
KI3: The company I work for identifies the employees' knowledge
KI4: The company I work for identifies the useful knowledge, which is available in the working process
KC
KC1: The company I work for has the mechanisms of creating and acquiring knowledge from different sources such as employee's and other companies
KC2: The company I work for encourages the lecturers to exchange their knowledge and ideas
KC3: The company I work for gives rewards for new ideas and knowledge
KC4: The company I work for has mechanisms for creating new knowledge from existing knowledge
KL
KL1: The company I work for collects the useful knowledge, which is identified from various sources
KL2: The company I work for gives the possibility to ask when there is a need to certain knowledge
KL3: The company I work for gives the information about what we know
KL4: The company I work for allows colleagues to ask about the abilities when there is a need to learn something
KO
KO1: The company I work for has a policy to review knowledge on a regular basis and keep them up to date
KO2: The company I work for has several mechanisms for filtering, cross-listing and integrating different types of knowledge
KO3: The company I work for gives feedback to lecturers about their ideas and knowledge
KO4: The company I work for has processes for applying knowledge learned from experiences and match sources of knowledge to problems and challenges
KS
KS1: The company I work for uses the databases, repositories and IT applications to store the knowledge for easy access by all lecturers
KS2: The company I work for uses various written devices such as newsletter, manuals to store the knowledge which capture from the employees
KS3: The company I work for has several publications to display the capture knowledge

(*continued*)

Table AI.
Questionnaire

	KS4: The company I work for has several mechanisms to store the knowledge for patent and copyright
	KD
	KD1: The company I work for has a form of knowledge that is readily accessible to employees who need it
	KD2: The company I work for sends out timely reports with appropriate information to employees
	KD3: The company I work for has libraries, resource center and other forums to display and distribute the knowledge
	KD4: The company I work for do lectures, conferences and training sessions to sharing knowledge
	KA
	KA1: The company I work for has several methods for lecturers to further develop their knowledge
	KA2: The company I work for has several mechanisms to protect the knowledge inside and outside the company
	KA3: The company I work for applies knowledge to critical competitive needs and quickly links sources of knowledge in problem-solving
	KA4: The company I work for has several methods to analyzing and evaluating knowledge to generate new patterns and knowledge for future use
	PS
	PS1: The project was implemented and used by the business
	PS2: The project was delivered within the allocated time
	PS3: The project was delivered within the budgeted cost
	PS4: The project was delivered within the agreed scope. Scope changes to be approved by the business
	PS5: The project achieved/realized the business expected commercial and user benefits as outlined in the business case
Table AI.	PS6: The project was delivered according to agreed quality

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